

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										817	USE ONLY FOR TIE IN
20	U04		MODULAR DBB	K06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
30	U04	VAW	MODULAR DBB	K06AC06	2	3	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
40	U04		MODULAR DBB	M06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
50	U04		MODULAR DBB	M06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	NT5	TO BE USED IN NON-PIGGED LINES ONLY
60	U04	VAW	MODULAR DBB	M06EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	3	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
70							***** BALL VALVES *****			UI2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
70		VB2									
80											
80	U04		BALL	A06AC05	2	4	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB LO	0	0		
90	U04		BALL	A06AC07	6	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0		
100	U04		BALL	A06AC06	2	4	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB LO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
110	U04		BALL	A06AC08	6	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		
120	U04		BALL	B06AC01	0.75	1.5	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
130	U04		BALL	B06AC02	0.75	1.5	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		
140	U04		BALL	B06AC05	2	3	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB LO	0	0		
150	U04	VA5	BALL	B06AC06	2	3	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB LO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
160	U04	VA5	BALL	B06AC07	4	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
170	U04	VA5	BALL	B06AC08	4	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
180	U04	817	BALL	B06CC07-K5E	26	28	BALL 600# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
190	U04	817	BALL	B06CC08-K5E	26	28	BALL 600# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
200							***** RISING STEM BALL VALVES *****				
200											
210		VA5	BALL	R06AC06	4	6	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0		
220		VA5	BALL	R06AC08	8	10	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0		
230							***** BUTTERFLY VALVES *****				
230											
240											
240			BUTTERFLY	T06AC61	10	24	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
250	U04	817	BUTTERFLY	T06CC61-K5E	26	28	BUTTERFLY 600# CF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
260							***** CHECK VALVES *****				
260											
270											
270	U04	VC1	CHECK	C06AC50	0.75	1.5	CHECK PISTON 600# RF LTCS A350 LF2 CL.1 TRIM 16	0	0		
280			CHECK	C06AC45	2	8	CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
290			CHECK	C06AC52	10	24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16	0	0		
300	U04	817	CHECK	C06CC52-K5E	26	26	CHECK DUAL DBL FLG 600# CF LTCS A352 LCC TRIM 16	0	0		
310							***** GLOBE VALVES *****				
310											
320											
320	U04		GLOBE	L06AC48	0.75	1.5	GLOBE 600# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0		
330			GLOBE	L06AC41	2	4	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
340			GLOBE	L06AC42	6	10	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
350							***** GATE VALVES *****				
350											
360											
360			GATE	W06AC70	3	4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
370			GATE	W06AC71	6	24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
380	U04	817	GATE	W06CC71-K5E	26	28	GATE 600# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
390							*****PIPE ABOVE GROUND*****				
390											
400											
400			PIPE	5358512	0.75	1.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0		
410			PIPE	5358515	2	2	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0		
420			PIPE	5358514	3	3	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
430											
430			PIPE	9001211	4	4	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE	0	0		
440			PIPE	9001255	6	6	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE	0	0		
450			PIPE	9001487	8	8	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE	0	0		
460			PIPE	9001488	10	12	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE	0	0		
470			PIPE	9001490	14	14	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE	0	0		
480			PIPE	9001495	16	16	LINEPIPE SMLS WT = 12.70 MM API-5L X60 BE	0	0		
490			PIPE	9001491	18	18	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE	0	0		
500			PIPE	9005241	20	20	LINEPIPE SMLS WT= 15.09 MM API-5L X60 BE	0	0		
510			PIPE	9005242	24	24	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND AND 26" 3GI FEED GAS UNDER GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 88 °C	12/16/2024	BY OZAC	U04	Page 1 of 7	600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
520			PIPE	9005243	26	26	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE	0	0		
530			PIPE	9005254	28	28	LINEPIPE SMLS WT= 20.62 MM API-5L X60 BE	0	0		
540		812	PIPE	5358511	0.5	0.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
550	U04		PIPE	9005244CTD	26	26	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE COATED	0	0		
560							***** PIPE BELOW GROUND *****				
570			PIPE	9001254CTD	4	6	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0		
580			PIPE	9001255CTD	6	6	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE COATED	0	0		
590			PIPE	9001487CTD	8	8	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED	0	0		
600			PIPE	9001488CTD	10	12	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE COATED	0	0		
610			PIPE	9001490CTD	14	14	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE COATED	0	0		
620			PIPE	9001495CTD	16	16	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE COATED	0	0		
630			PIPE	9001491CTD	18	18	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE COATED	0	0		
640			PIPE	9005241CTD	20	20	LINEPIPE SMLS WT= 15.09 MM API-5L X60 BE COATED	0	0		
650			PIPE	9005242CTD	24	28	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE COATED	0	0		
660							***** FITTINGS ABOVE & BELOW GROUND *****				
670											
680			ELBOW	5392816	0.75	2	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0		
690			ELBOW	5392600	3	3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
690			ELBOW	9002190	4	4	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60	0	0		
700			ELBOW	9006470	6	6	ELL BW 90 DEG LR WT 7.11 MM A860 WPHY 60	0	0		
710			ELBOW	9002157	8	8	ELL BW 90 DEG LR WT 8.74MM A860 WPHY 60	0	0		
720			ELBOW	9006468	10	12	ELL BW 90 DEG LR WT 11.13 MM A860 WPHY 60	0	0		
730			ELBOW	9006479	16	16	ELL BW 90 DEG LR WT 12.70 MM A860 WPHY 60	0	0		
740			ELBOW	9006480	18	18	ELL BW 90 DEG LR WT 14.27 MM A860 WPHY 60	0	0		
750			ELBOW	9006482	20	20	ELL BW 90 DEG LR WT 15.09 MM A860 WPHY 60-W	0	0		
760			ELBOW	9006474	24	24	ELL BW 90 DEG LR WT 17.48 MM A860 WPHY 60-W	0	0		
770	U04		ELBOW	9006568	26	26	ELL BW 90 DEG LR WT 19.05 MM A860 WPHY 60	0	0		
780			ELBOW	5392822	0.75	2	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0		
790			ELBOW	5392606	3	3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
800	U04		ELBOW	9006574	6	6	ELL BW 45 DEG LR WT 7.11MM A860 WPHY 60	0	0		
810			ELBOW	9006569	12	12	ELL BW 45 DEG LR WT 11.13MM A860 WPHY 60	0	0		
820	U04		ELBOW	9006567	26	26	ELL BW 45 DEG LR WT 19.05MM A860 WPHY 60	0	0		
830											
830			TEE	5392825	0.75	2	TEE BW SCH 160 LTCS A420 WPL6	0	0		
840			TEE	5392609	3	3	TEE BW SCH XS LTCS A420 WPL6	0	0		
850		NT5	TEE	9301609	4	4	TEE BW WT 6.35 MM LTCS A860 WPHY-60	0	0		
860		NT5	TEE	9006600	6	6	TEE BW WT 7.11MM A860 WPHY 60	0	0		
870		NT5	TEE	9301607	8	8	TEE BW WT 8.74 MM LTCS A860 WPHY-60	0	0		
880		NT5	TEE	9006604	10	12	TEE BW WT 11.13MM A860 WPHY 60	0	0		
890		NT5	TEE	9006632	14	14	TEE BW WT 11.91MM A860 WPHY 60	0	0		
900		NT5	TEE	9006633	16	16	TEE BW WT 12.7 MM A860 WPHY 60	0	0		
910		NT5	TEE	9006486	18	18	TEE BW WT 14.27MM A860 WPHY 60	0	0		
920		NT5	TEE	9006488	20	20	TEE BW WT 15.09MM A860 WPHY 60-W	0	0		
930		NT5	TEE	9006489	24	24	TEE BW WT 17.48MM A860 WPHY 60-W	0	0		
940	U04	NT5	TEE	9006666	26	26	TEE BW WT 19.05MM A860 WPHY 60	0	0		
950			TEE RED	5392821	0.75	1.5	TEE RED BW SCH 160 LTCS A420 WPL6	1	2		
960			TEE RED	5392469	2	2	TEE RED BW SCH XS/160 LTCS A420 WPL6	3	3		
970		NT5	TEE RED	9006608	2	3	TEE RED BW WT 6.35 MM / SCH XS LTCS A860 WPHY 60	4	4		
980		NT5	TEE RED	9006609	3	3	TEE RED BW WT 7.11 MM/SCH XS LTCS A860 WPHY 60	6	6		
990		NT5	TEE RED	9004208	4	4	TEE RED BW WT 7.11 MM/6.35 MM LTCS A860 WPHY 60	6	6		
1000		NT5	TEE RED	9004123	4	4	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60	8	8		
1010		NT5	TEE RED	9006611	6	6	TEE RED BW WT 8.74 MM/7.11 MM LTCS A860 WPHY 60	8	8		
1020		NT5	TEE RED	9006612	4	4	TEE RED BW WT 11.13 MM/6.35 MM LTCS A860 WPHY 60	10	10		
1030		NT5	TEE RED	9004209	6	6	TEE RED BW WT 11.13 MM/7.11 MM LTCS A860 WPHY 60	10	12		
1040		NT5	TEE RED	9006485	8	8	TEE RED BW WT 11.13 MM/8.74 MM LTCS A860 WPHY 60	10	12		
1050		NT5	TEE RED	9006613	10	10	TEE RED BW WT 11.13 MM LTCS A860 WPHY 60	12	12		
1060		NT5	TEE RED	9006028	6	6	TEE RED BW WT 11.91 MM/7.11 MM LTCS A860 WPHY 60	14	14		
1070		NT5	TEE RED	9006614	8	8	TEE RED BW WT 11.91 MM/8.74 MM LTCS A860 WPHY 60	14	14		
1080		NT5	TEE RED	9006615	10	12	TEE RED BW WT 11.91 MM/11.13 MM LTCS A860 WPHY 60	14	14		
1090			TEE RED	9006602	6	6	TEE RED BW WT 12.70 MM/7.11 MM LTCS A860 WPHY 60	16	16		
1100		NT5	TEE RED	9006616	8	8	TEE RED BW WT 12.70 MM/8.74 MM LTCS A860 WPHY 60	16	16		
1110		NT5	TEE RED	9006617	10	12	TEE RED BW WT 12.70 MM/11.13 MM LTCS A860 WPHY 60	16	16		
1120		NT5	TEE RED	9006618	14	14	TEE RED BW WT 12.70 MM/11.91 MM LTCS A860 WPHY 60	16	16		
1130		NT5	TEE RED	9006619	8	8	TEE RED BW WT 14.27 MM/8.74 MM LTCS A860 WPHY 60-W	18	18		
1140		NT5	TEE RED	9006620	10	12	TEE RED BW WT 14.27 MM/11.13 MM LTCS A860 WPHY 60-W	18	18		
1150		NT5	TEE RED	9006621	14	14	TEE RED BW WT 14.27 MM/11.91 MM LTCS A860 WPHY 60-W	18	18		
1160		NT5	TEE RED	9006622	16	16	TEE RED BW WT 14.27 MM/12.7 MM LTCS A860 WPHY 60-W	18	18		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND AND 26" 3GI FEED GAS UNDER GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 88 °C	12/16/2024	BY OZAC	U04	Page 2 of 7	600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
1170			TEE RED	9006603	8	8	TEE RED BW WT 15.09 MM/8.74 MM LTCS A860 WPHY 60-W	20	20		
1180		NT5	TEE RED	9006623	10	12	TEE RED BW WT 15.09 MM/11.13 MM LTCS A860 WPHY 60-W	20	20		
1190		NT5	TEE RED	9006624	14	14	TEE RED BW WT 15.09 MM/11.91 MM LTCS A860 WPHY 60-W	20	20		
1200		NT5	TEE RED	9006625	16	16	TEE RED BW WT 15.09 MM/12.7 MM LTCS A860 WPHY 60-W	20	20		
1210		NT5	TEE RED	9006626	18	18	TEE RED BW WT 15.09 MM/14.27 MM LTCS A860 WPHY 60-W	20	20		
1220		NT5	TEE RED	9006627	10	12	TEE RED BW WT 17.48 MM/11.13 MM LTCS A860 WPHY 60-W	24	24		
1230		NT5	TEE RED	9006628	14	14	TEE RED BW WT 17.48 MM/11.91 MM LTCS A860 WPHY 60-W	24	24		
1240		NT5	TEE RED	9006629	16	16	TEE RED BW WT 17.48 MM/12.7 MM LTCS A860 WPHY 60-W	24	24		
1250		NT5	TEE RED	9006630	18	18	TEE RED BW WT 17.48 MM/14.27 MM LTCS A860 WPHY 60-W	24	24		
1260		NT5	TEE RED	9006631	20	20	TEE RED BW WT 17.48 MM/15.09 MM LTCS A860 WPHY 60-W	24	24		
1270											
1270			CAP	5392826	0.75	2	CAP BW SCH 160 LTCS A420 WPL6	0	0		
1280			CAP	5392610	3	3	CAP BW SCH XS LTCS A420 WPL6	0	0		
1290			CAP	9005071	4	4	CAP BW WT 6.35 MM LTCS A860 WPHY-60	0	0		
1300			CAP	9005056	6	6	CAP BW WT 7.11MM LTCS A860 WPHY-60	0	0		
1310			CAP	9005059	8	8	CAP BW WT 8.74 MM LTCS A860 WPHY-60	0	0		
1320			CAP	9005064	10	12	CAP BW WT 11.13 MM LTCS A860 WPHY-60	0	0		
1330			CAP	9005065	14	14	CAP BW WT 11.91 MM LTCS A860 WPHY-60	0	0		
1340			CAP	9005067	16	16	CAP BW WT 12.70 MM LTCS A860 WPHY-60	0	0		
1350			CAP	9005068	18	18	CAP BW WT 14.27 MM LTCS A860 WPHY-60	0	0		
1360			CAP	9005091	20	20	CAP BW WT 15.09 MM LTCS A860 WPHY-60	0	0		
1370			CAP	9005070	24	24	CAP BW WT 17.48 MM LTCS A860 WPHY-60	0	0		
1380	U04		CAP	9005072	26	26	CAP BW WT 19.05 MM LTCS A860 WPHY-60	0	0		
1390			CAP	9005073	28	28	CAP BW WT 20.62 MM LTCS A860 WPHY-60	0	0		
1400											
1400			RED CONC	5392827	0.75	1.5	REDUCER CONC SCH 160 LTCS A420 WPL6	1	2		
1410			RED CONC	5392462	2	2	REDUCER CONC SCH XS/160 LTCS A420 WPL6	3	3		
1420			RED ECC	5392828	0.75	1.5	REDUCER ECC SCH 160 LTCS A420 WPL6	1	2		
1430			RED ECC	5392465	2	2	REDUCER ECC SCH XS/160 LTCS A420 WPL6	3	3		
1440			RED CONC	9006923	1.5	2	REDUCER CONC WT 6.35MM / SCH 160 A860 WPHY 60	4	4		
1450			RED CONC	9006919	3	3	REDUCER CONC WT 6.35 MM / XS WT LTCS A860 WPHY-60	4	4		
1460			RED CONC	9006920	3	3	REDUCER CONC WT 7.11 MM / SCH XS MM LTCS A860 WPHY-60	6	6		
1470	U04		RED CONC	9006904	4	4	REDUCER CONC WT 7.11 MM / 6.35 MM LTCS A860 WPHY-60	6	6		
1480	U04		RED CONC	9002348	6	6	REDUCER CONC WT 8.74 / WT 7.11 MM A860 WPHY 60	8	8		
1490			RED ECC	9002042	1.5	2	REDUCER ECC WT 6.35 MM / SCH 160 LTCS A860 WPHY-60	4	4		
1500			RED ECC	9001581	3	3	REDUCER ECC WT 6.35 MM / XS WT LTCS A860 WPHY-60	4	4		
1510			RED ECC	9001582	3	3	REDUCER ECC WT 7.11 MM / SCH XS MM LTCS A860 WPHY-60	6	6		
1520			RED ECC	9001583	4	4	REDUCER ECC WT 7.11 MM / 6.35 MM LTCS A860 WPHY-60	6	6		
1530			RED ECC	9002292	4	4	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	8	8		
1540			RED ECC	9001377	6	6	REDUCER ECC WT 8.74 MM / 7.11 MM LTCS A860 WPHY-60	8	8		
1550			RED ECC	9001586	4	4	REDUCER ECC WT 11.13 MM / 6.35MM LTCS A860 WPHY-60	10	10		
1560			RED ECC	9002021	6	6	REDUCER ECC WT 11.13MM / 7.11MM LTCS A860 WPHY-60	10	12		
1570			RED ECC	9002402	8	8	REDUCER ECC WT 11.13 MM / 8.74 MM LTCS A860 WPHY-60	10	12		
1580			RED ECC	9002022	10	10	REDUCER ECC WT 11.13MM LTCS A860 WPHY-60	12	12		
1590			RED ECC	9002023	6	6	REDUCER ECC WT 11.91MM / 7.11MM LTCS A860 WPHY-60	14	14		
1600			RED ECC	9001380	8	8	REDUCER ECC WT 11.91MM / 8.74MM LTCS A860 WPHY-60	14	14		
1610			RED ECC	9001393	10	12	REDUCER ECC WT 11.91MM / 11.13MM LTCS A860 WPHY-60	14	14		
1620			RED ECC	9005116	8	8	REDUCER ECC WT 12.70 MM / 8.74 MM LTCS A860 WPHY-60	16	16		
1630			RED ECC	9002407	10	12	REDUCER ECC WT 12.70 MM / 11.13 MM LTCS A860 WPHY-60	16	16		
1640			RED ECC	9001590	14	14	REDUCER ECC WT 12.70 MM / 11.91 MM LTCS A860 WPHY-60	16	16		
1650			RED ECC	9002409	10	12	REDUCER ECC WT 14.27 MM / 11.13 MM LTCS A860 WPHY-60	18	18		
1660			RED ECC	9001591	14	14	REDUCER ECC WT 14.27 MM / 11.91 MM LTCS A860 WPHY-60	18	18		
1670			RED ECC	9002410	16	16	REDUCER ECC WT 14.27 MM / 12.70 MM LTCS A860 WPHY-60	18	18		
1680			RED ECC	9006301	12	12	REDUCER ECC WT 15.09 MM / 11.13MM LTCS A860 WPHY-60-W	20	20		
1690			RED ECC	9001593	14	14	REDUCER ECC WT 15.09 MM / 11.91 MM LTCS A860 WPHY-60-W	20	20		
1700			RED ECC	9001594	16	16	REDUCER ECC WT 15.09 MM / 12.70 MM LTCS A860 WPHY-60-W	20	20		
1710			RED ECC	9001595	18	18	REDUCER ECC WT 15.09 MM / 14.27 MM LTCS A860 WPHY-60-W	20	20		
1720	U04		RED ECC	9001592	12	12	REDUCER ECC WT 15.09 MM / 11.13MM LTCS A860 WPHY-60	20	20		
1730	U04		RED ECC	9006259	18	18	REDUCER ECC WT 15.09 MM / 14.27 MM LTCS A860 WPHY-60	20	20		
1740	U04		RED ECC	9001388	18	18	REDUCER ECC WT 17.48 MM / 14.27MM LTCS A860 WPHY-60-W	24	24		
1750			RED ECC	9001387	16	16	REDUCER ECC WT 17.48 MM / 12.70 MM LTCS A860 WPHY-60-W	24	24		
1760	U04		RED ECC	9001596	20	20	REDUCER ECC WT 17.48 MM / 15.09 MM LTCS A860 WPHY-60-W	24	24		
1770			RED ECC	9001596	20	20	REDUCER ECC WT 17.48 MM / 15.09 MM LTCS A860 WPHY-60-W	24	24		
1780			RED ECC	9002024	18	18	REDUCER ECC WT 20.62MM / 14.27MM LTCS A860 WPHY-60-W	28	28		
1790			RED ECC	9001597	20	20	REDUCER ECC WT 20.62MM / 15.09 MM LTCS A860 WPHY-60-W	28	28		
1800			RED ECC	9001598	24	24	REDUCER ECC WT 20.62MM / 17.48 MM LTCS A860 WPHY-60-W	28	28		
1810											
1810							***** INDUCTION BENDS ABOVE GROUND*****				
1820											
1820			INDUC BEND	9001534	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1830			INDUC BEND	9001396	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1840			INDUC BEND	9001816	8	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND AND 26" 3GI FEED GAS UNDER GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: - 40 / 88 °C	12/16/2024	BY OZAC	U04	Page 3 of 7	600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1850			INDUC BEND	9001899	10	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1860			INDUC BEND	9001397	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1870			INDUC BEND	9005321	16	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1880			INDUC BEND	9005322	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1890			INDUC BEND	9001572	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1900			INDUC BEND	9001573	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1910			INDUC BEND	9001574	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1920			INDUC BEND	9001575	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1930			INDUC BEND	9001399	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1940			INDUC BEND	9001826	8	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1950			INDUC BEND	9005326	10	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1960			INDUC BEND	9001500	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1970			INDUC BEND	9005327	16	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1980			INDUC BEND	9005328	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1990			INDUC BEND	9001577	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2000			INDUC BEND	9005040	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2010			INDUC BEND	9002026	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2020			INDUC BEND	9001580	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2030			INDUC BEND	9002000	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2040			INDUC BEND	9001836	8	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2050			INDUC BEND	9002001	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2060			INDUC BEND	9005333	16	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2070			INDUC BEND	9005334	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2080			INDUC BEND	9002811	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2090			INDUC BEND	9001579	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2100			INDUC BEND	9002062	26	26	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2110			INDUC BEND	9002027	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 20.62MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2120							***** INDUCTION BENDS BELOW GROUND*****			
2130										
2130			INDUC BEND	9001534CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2140			INDUC BEND	9001396CTD	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2150			INDUC BEND	9001816CTD	8	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2160			INDUC BEND	9001899CTD	10	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2170			INDUC BEND	9001397CTD	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2180			INDUC BEND	9005321CTD	16	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2190			INDUC BEND	9005322CTD	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2200			INDUC BEND	9001572CTD	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2210			INDUC BEND	9001573CTD	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2220			INDUC BEND	9001574CTD	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2230	U04		INDUC BEND	9007038CTD	12	12	INDUCTION BEND 90 DEG X 10D RADIUS WT 11.13MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2240	U04		INDUC BEND	9007039CTD	18	18	INDUCTION BEND 90 DEG X 10D RADIUS WT 14.27MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2250	U04		INDUC BEND	9007032CTD	6	6	INDUCTION BEND 90 DEG X 10D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2260	U04		INDUC BEND	9007047CTD	12	12	INDUCTION BEND 90 DEG X 15D RADIUS WT 11.13MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2270	U04		INDUC BEND	9007044CTD	6	6	INDUCTION BEND 90 DEG X 15D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2280			INDUC BEND	9001575CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2290			INDUC BEND	9001399CTD	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2300			INDUC BEND	9001826CTD	8	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2310			INDUC BEND	9005326CTD	10	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2320			INDUC BEND	9004001CTD	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2330			INDUC BEND	9005327CTD	16	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2340			INDUC BEND	9005328CTD	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2350			INDUC BEND	9001577CTD	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2360			INDUC BEND	9005040CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND AND 26" 3GI FEED GAS UNDER GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCs FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 88 °C	12/16/2024	BY OZAC	U04	Page 4 of 7	600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
2370			INDUC BEND	9002026CTD	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2380	U04		INDUC BEND	9007009CTD	12	12	INDUCTION BEND 45DEG X 10D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2390	U04		INDUC BEND	9007010CTD	18	18	INDUCTION BEND 45DEG X 10D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2400	U04		INDUC BEND	9007011CTD	20	20	INDUCTION BEND 45DEG X 10D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2410	U04		INDUC BEND	9007013CTD	24	24	INDUCTION BEND 45DEG X 10D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2420	U04		INDUC BEND	9007002CTD	6	6	INDUCTION BEND 45DEG X 10D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2430	U04		INDUC BEND	9007024CTD	12	12	INDUCTION BEND 45DEG X 15D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2440	U04		INDUC BEND	9007025CTD	18	18	INDUCTION BEND 45DEG X 15D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2450	U04		INDUC BEND	9007026CTD	20	20	INDUCTION BEND 45DEG X 15D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2460	U04		INDUC BEND	9007028CTD	24	24	INDUCTION BEND 45DEG X 15D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2470	U04		INDUC BEND	9007017CTD	6	6	INDUCTION BEND 45DEG X 15D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2480			INDUC BEND	9001580CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2490			INDUC BEND	9002000CTD	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2500			INDUC BEND	9001836CTD	8	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2510			INDUC BEND	9005331CTD	10	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2520			INDUC BEND	9002001CTD	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2530			INDUC BEND	9004000CTD	16	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2540			INDUC BEND	9005334CTD	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2550			INDUC BEND	9002811CTD	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2560			INDUC BEND	9001579CTD	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2570			INDUC BEND	9002062CTD	26	26	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2580	U04		INDUC BEND	9002002CTD	26	26	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2590			INDUC BEND	9002027CTD	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 20.62MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2600							***** OLETS ABOVE & BELOW GROUND*****			
2600										
2610	U04		OLETS	5505392	0.75	1	WELDOLET SCH 160 LTCS A350 LF2 CL.1	2	2	
2620	U04		OLETS	5505393	0.75	1	WELDOLET SCH XS/160 LTCS A350 LF2 CL.1	3	3	
2630			OLETS	9002450	0.75	1.5	WELDOLET WT 6.35 MM/SCH 160 WT LTCS A694 F60	4	4	
2640			OLETS	9002072	0.75	2	WELDOLET WT 7.11MM/ SCH 160 LTCS A694 F60	6	6	
2650			OLETS	9002455	0.75	2	WELDOLET WT 8.74MM/ SCH 160 LTCS A694 F60	8	8	
2660			OLETS	9006423	3	3	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	8	8	
2670			OLETS	9002076	0.75	2	WELDOLET WT 11.13MM/ SCH 160 LTCS A694 F60	10	12	
2680			OLETS	9005119	3	3	WELDOLET WT 11.13 MM/ SCH XS WT LTCS A694 F60	10	12	
2690			OLETS	9005120	4	4	WELDOLET WT 11.13 MM/ 6.35 MM WT LTCS A694 F60	12	12	
2700			OLETS	9002080	0.75	2	WELDOLET WT 11.91MM/ SCH 160 LTCS A694 F60	14	14	
2710			OLETS	9002003	3	3	WELDOLET WT 11.91MM/SCH XS LTCS A694 F60	14	14	
2720			OLETS	9002009	4	4	WELDOLET WT 11.91MM/6.35MM LTCS A694 F60	14	14	
2730			OLETS	9002466	0.75	2	WELDOLET WT 12.70 MM/ SCH 160 WT LTCS A694 F60	16	16	
2740			OLETS	9002467	3	3	WELDOLET WT 12.70 MM/ SCH XS WT LTCS A694 F60	16	16	
2750			OLETS	9002469	4	4	WELDOLET WT 12.70 MM/ 6.35 MM WT LTCS A694 F60	16	16	
2760			OLETS	9002470	0.75	2	WELDOLET WT 14.27 MM/ SCH 160 WT LTCS A694 F60	18	18	
2770			OLETS	9002471	3	3	WELDOLET WT 14.27 MM/ SCH XS WT LTCS A694 F60	18	18	
2780			OLETS	9002473	4	4	WELDOLET WT 14.27 MM/ 6.35 MM LTCS A694 F60	18	18	
2790			OLETS	9005121	6	6	WELDOLET WT 14.27 MM/ 7.11 MM LTCS A694 F60	18	18	
2800			OLETS	9002820	0.75	2	WELDOLET WT 15.09 MM/ SCH 160 LTCS A694 F60	20	20	
2810			OLETS	9005124	3	3	WELDOLET WT 15.09 MM/ SCH XS WT LTCS A694 F60	20	20	
2820			OLETS	9005122	4	4	WELDOLET WT 15.09 MM/ 6.35 MM LTCS A694 F60	20	20	
2830			OLETS	9005123	6	6	WELDOLET WT 15.09 MM/ 7.11 MM LTCS A694 F60	20	20	
2840			OLETS	9004336	0.75	2	WELDOLET WT 17.48 MM/SCH 160 LTCS A694 F60	24	24	
2850			OLETS	9005053	3	3	WELDOLET WT 17.48 MM/SCH XS MM LTCS A694 F60	24	24	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND AND 26" 3GI FEED GAS UNDER GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: - 40 / 88 °C	12/16/2024	BY OZAC	U04	Page 5 of 7	600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
2860			OLETS	9005049	4	4	WELDOLET WT 17.48 MM/6.35 MM LTCS A694 F60	24	24		
2870			OLETS	9005050	6	6	WELDOLET WT 17.48 MM/7.11 MM LTCS A694 F60	24	24		
2880			OLETS	9005052	8	8	WELDOLET WT 17.48 MM/8.74 MM LTCS A694 F60	24	24		
2890			OLETS	9002089	0.75	2	WELDOLET WT 19.05 MM/ SCH 160 LTCS A694 F60	26	26		
2900			OLETS	9006252	3	3	WELDOLET WT 19.05 MM/ SCH XS LTCS A694 F60	26	26		
2910	U04		OLETS	9002092	4	4	WELDOLET WT 19.05 MM/ 6.35 MM LTCS A694 F60	26	26		
2920	U04		OLETS	9002093	6	6	WELDOLET WT 19.05 MM/ 7.11 MM LTCS A694 F60	26	26		
2930	U04		OLETS	9002090	8	8	WELDOLET WT 19.05 MM/ 8.74 MM LTCS A694 F60	26	26		
2940			OLETS	9002028	4	4	WELDOLET WT 20.62 MM/ 6.35 MM LTCS A694 F60	28	28		
2950			OLETS	9002029	6	6	WELDOLET WT 20.62 MM/ 7.11 MM LTCS A694 F60	28	28		
2960			OLETS	9002095	8	8	WELDOLET WT 20.62 MM/ 8.74 MM LTCS A694 F60	28	28		
2970			OLETS	9002030	10	10	WELDOLET WT 20.62 MM/ 11.13 MM LTCS A694 F60	28	28		
2980											
2980	U04		OLETS	5504460	0.75	1.5	NIPOFLANGE 600# RF SCH 160 LTCS A350 LF2 CL.1	2	2		
2990	U04		OLETS	5504461	0.75	2	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2 CL.1	3	3		
3000			OLETS	9002070	0.75	2	NIPOFLANGE 600# RF WT 6.35MM/SCH 160 LTCS A694 F60	4	4		
3010			OLETS	9002065	0.75	2	NIPOFLANGE 600# RF WT 7.11 MM/SCH 160 LTCS A694 F60	6	6		
3020			OLETS	9003454	0.75	2	NIPOFLANGE 600# RF WT 8.74 MM/SCH 160 LTCS A694 F60	8	8		
3030			OLETS	9003457	0.75	2	NIPOFLANGE 600# RF WT 11.13 MM/SCH 160 WT LTCS A694 F60	10	12		
3040			OLETS	9002066	0.75	2	NIPOFLANGE 600# RF WT 11.91MM/SCH 160 LTCS A694 F60	14	14		
3050			OLETS	9003458	0.75	2	NIPOFLANGE 600# RF WT 12.70 MM/SCH 160 WT LTCS A694 F60	16	16		
3060			OLETS	9003459	0.75	2	NIPOFLANGE 600# RF WT 14.27 MM/SCH 160 WT LTCS A694 F60	18	18		
3070			OLETS	9006495	0.75	2	NIPOFLANGE 600# RF 15.09 MM/SCH 160 LTCS A694 F60	20	20		
3080			OLETS	9006496	0.75	2	NIPOFLANGE 600# RF 17.48 MM/SCH 160 LTCS A694 F60	24	24		
3090			OLETS	9002069	0.75	2	NIPOFLANGE 600# RF WT 20.62MM/SCH 160 LTCS A694 F60	28	28		
3100	U04		OLETS	9002068	0.75	2	NIPOFLANGE 600# RF WT 19.05MM/SCH 160 LTCS A694 F60	26	26		
3110	U04	UJ2	OLETS	9008203	1.5	1.5	NIPOFLANGE 600# RF WT 19.05MM/SCH 160 LTCS A694 F60	26	26		
3120											
3120							***** FLANGES ABOVE & BELOW GROUND*****				
3130											
3130			FLANGE	5851204	0.5	2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE	0	0		
3140											
3140	U04		FLANGE	5557038	0.5	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 160	0	0		
3150	U04		FLANGE	5557036	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
3160			FLANGE	9001471	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.35MM	0	0		
3170			FLANGE	9001462	6	6	FLG WN 600# RF LTCS A694 F60 WT 7.11MM	0	0		
3180			FLANGE	9003484	8	8	FLG WN 600# RF LTCS A694 F60 WT 8.74 MM	0	0		
3190			FLANGE	9001464	10	12	FLG WN 600# RF LTCS A694 F60 WT 11.13MM	0	0		
3200			FLANGE	9001511	14	14	FLG WN 600# RF LTCS A694 F60 WT 11.91MM	0	0		
3210			FLANGE	9001512	16	16	FLG WN 600# RF LTCS A694 F60 WT 12.70MM	0	0		
3220			FLANGE	9001515	18	18	FLG WN 600# RF LTCS A694 F60 WT 14.27MM	0	0		
3230			FLANGE	9001514	20	20	FLG WN 600# RF LTCS A694 F60 WT 15.09MM	0	0		
3240			FLANGE	9001513	24	24	FLG WN 600# RF LTCS A694 F60 WT 17.48MM	0	0		
3250			FLANGE	9003496	28	28	FLG WN 600# RF LTCS A694 F60 WT 20.62 MM SERIES A	0	0		
3260	U04	817	FLANGE	9006169	26	26	FLG COMPACT 600# RJ A694 F60 WT 19.05 MM	0	0		
3270	U04	817	FLANGE	9001756	28	28	FLG COMPACT 600# RJ A694 F60 WT 20.62 MM	0	0		
3280	U04		BLIND	5615598	0.5	3	FLG BLIND 600# RF LTCS A350 LF2 CL.1	0	0		
3290			BLIND	9001518	4	24	FLG BLIND 600# RF LTCS A694 F60	0	0		
3300	U04		FLANGE	9003498	26	26	FLG WN 600# RF LTCS A694 F60 WT 19.05 MM SERIES A	0	0		
3310			BLIND	9001640	28	28	FLG BLIND 600# RF LTCS A694 F60 B16.47 SERIES A	0	0		
3320	U04	817	BLIND	9619769	26	28	FLG COMPACT BLIND 600# RJ LTCS A694 F60 PIPE CLASS 600K5E	0	0		
3330	U04		TEE RED	5599705	0.5	0.5	FLG SCRDR (FNPT) REDUCING 600# RF A350 LF2 CL.1	0.75	1.5		
3340	U04	817	FLANGE	9006704	4	4	FLG WN 600# RTJ A694 F60 WT 6.35MM	0	0		
3350	U04	817	FLANGE	9006705	6	6	FLG WN 600# RTJ A694 F60 WT 7.11MM	0	0		
3360	U04	817	FLANGE	9006720	8	8	FLG WN 600# RTJ A694 F60 WT 8.74MM	0	0		
3370	U04	817	FLANGE	9006706	12	12	FLG WN 600# RTJ A694 F60 WT 11.13MM	0	0		
3380	U04	817	FLANGE	9006708	20	20	FLG WN 600# RTJ A694 F60 WT 15.09MM	0	0		
3390	U04		FLANGE	9002847	8	8	FLG WN 900# RF LTCS A694 F60 WT 8.74MM	0	0		
3400											
3400							*****GASKETS*****				
3410											
3410			GASKET	5673103	0.5	24	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0		
3420	U04	817	GASKET	5673416	26	28	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 600K5E	0	0		
3430		SB6	GASKET	5674012	26	28	GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0		
3440	U04	817	GASKET	5673202	4	20	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0		
3450											
3450		B1					***** BOLTING *****				
3460											
3460	U04		BOLTING	5675770	0.5	28	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0		
3470											
3470							***** BLINDS & SPACERS *****				
3480											

	TENGIZCHEVROI TCO TENGI, KAZAKHSTAN	CONSTRUCTION		CORR.ALLOW.		SERVICE		MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND AND 26" 3GI FEED GAS UNDER GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 3 MM		SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO		MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 88 °C		12/16/2024	BY OZAC	U04	Page 6 of 7	600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
3480			SPEC BLIND	5749587	0.75	8	SPECTACLE BLIND 600# RF LTCS A516-GR70	0	0	
3490										
3490			PADDLE	5750040	10	24	PADDLE BLANK 600# RF LTCS A516-GR70	0	0	
3500			PADDLE	5750030	28	28	PADDLE BLANK 600# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
3510			PADDLE	5750251	10	24	PADDLE SPACER 600# RF LTCS A516-GR70	0	0	
3520			PADDLE	5750252	28	28	PADDLE SPACER 600# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
3530	U04	817	PADDLE	5750214	26	28	PADDLE BLANK COMPACT 600# RJ LTCS A516 GR.70	0	0	
3540	U04	817	PADDLE	5750236	26	28	PADDLE SPACER COMPACT 600# RJ LTCS A516 GR.70	0	0	
3550										
3550	U04		GASKET	5672251	2	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3560	U04		GASKET	5673117	3	8	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3570										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICES ALL WELDS ABOVE GROUND AND 26" 3GI FEED GAS UNDER GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 88 °C	12/16/2024	BY OZAC	U04	Page 7 of 7	600K5E